CodeAlpha Robotics & Automation Internship

Task 3: Automation System Design

Project: Smart Home Lighting Automation System

Objective

Design an automated lighting system that turns lights ON when motion is detected and OFF after a period of inactivity. The system improves energy efficiency and user convenience.

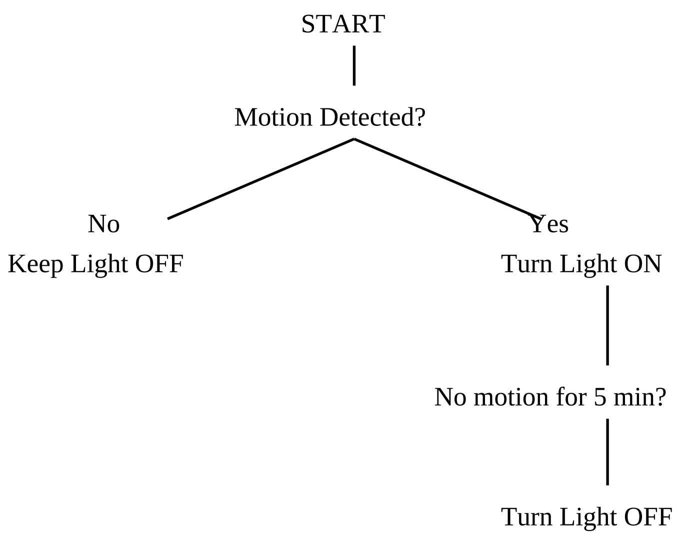
Components

Component	Purpose
PIR Motion Sensor	Detects human movement
Arduino Uno	Processes sensor input
Relay Module	Controls AC light safely
LED/Bulb	Lighting load
Power Supply	Provides electrical power

Working Principle

1. PIR sensor continuously monitors motion.
2. If motion is detected, Arduino activates the relay.
3. Relay switches the light ON.
4. If no motion is detected for a preset time (e.g., 5 minutes), Arduino deactivates the relay and turns the light OFF.
5. The cycle repeats continuously.

Flowchart



Applications

Homes, offices, classrooms, washrooms, parking areas, corridors, warehouses, and smart buildings.

Advantages

- Saves electricity
- Improves convenience
- Extends bulb life
- Reduces manual operation
- Supports smart-home automation

Limitations

False triggering due to pets or heat sources, sensor range limitations, and dependency on electrical power.

Conclusion

The Smart Home Lighting Automation System is a simple and cost-effective automation project that demonstrates the integration of sensors, controllers, and actuators. It improves energy efficiency while providing a practical introduction to robotics and automation concepts.